ELEC 271

Section 3 - Implementation Technology

Fall, 2026

Instructor: Dr. Muhammad Saad Asad

Modified from notes provided by Prof. Ryan Grant

Copyright © Prof. Ryan Grant

Lecture Outline

- Logic Values and Voltage Levels, Noise Margin
- Transistor Switches
- NMOS and CMOS Logic Gates
- Standard Chips, Programmable Logic Devices
    - PLAs, PALs, CPLDs, FPGAs
- Custom Chips, Standard Cells, Gate Arrays
- Propagation Delay, Fan-in and Fan-out, Buffers, Tri-state Buffers
- Exclusive-OR Gates
-

Read Chapter 3 of the textbook

Implementation technology

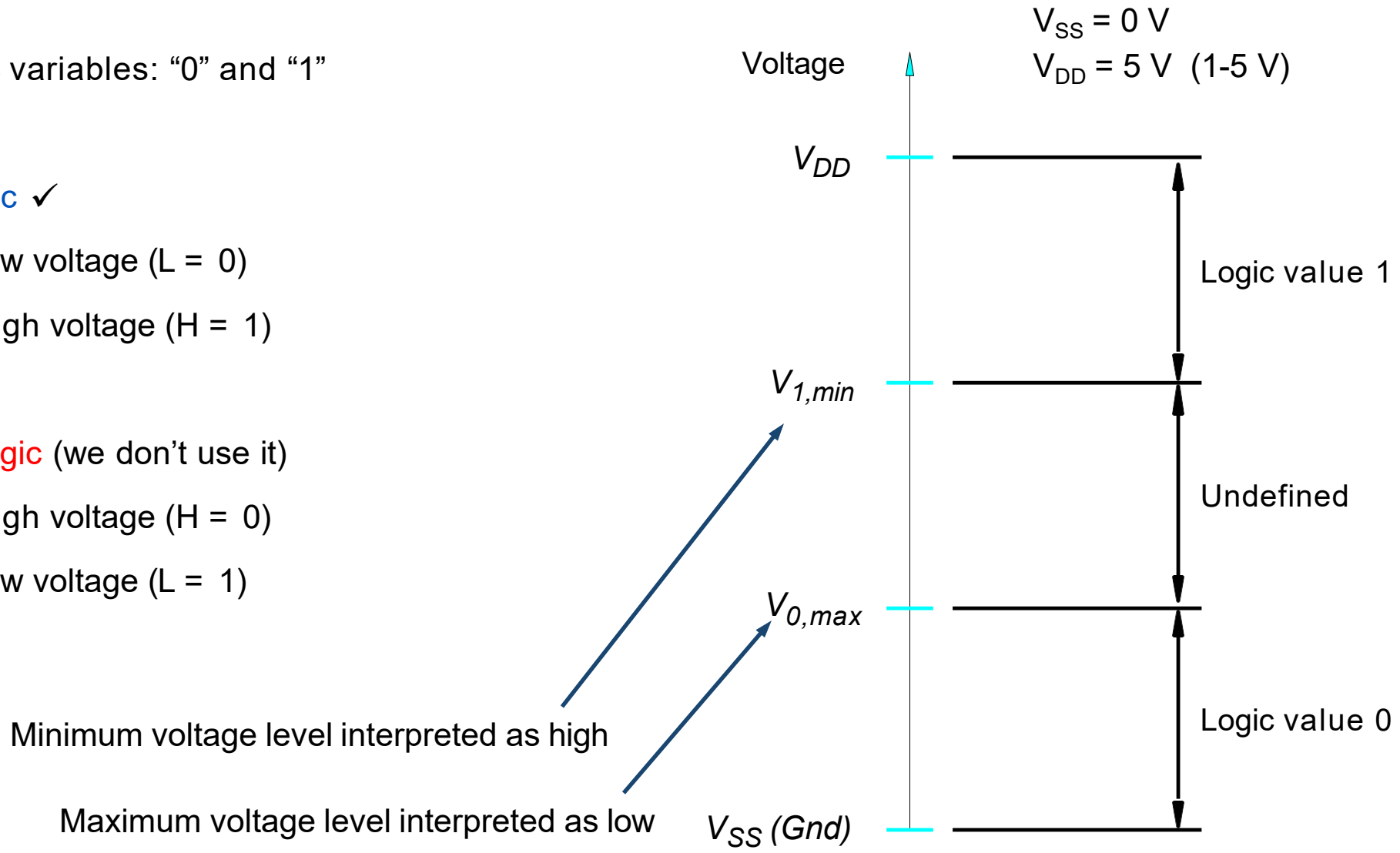
- This chapter will have a detailed discussion of integrated circuit technology. We will show how transistors operate as simple switches, and how logic gates are built using transistors. We also describe the structure of integrated circuit chips that are used to implement logic circuits.

- We will start by first understanding how logic variables can be physically represented as signals in electronic circuits. Our discussion will be restricted to binary variables, which can take on only the values 0 and 1. In a circuit these values can be represented either as levels of voltage or current. Both alternatives are used in different technologies. We will focus on the simplest and most popular representation, using voltage levels.

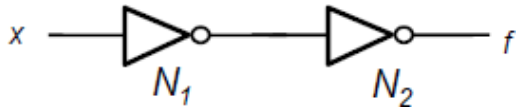
Logic Values and Voltage Levels

- Binary logic variables: “0” and “1”
- Positive logic ✓
  - “0” – low voltage (L = 0)
  - “1” – high voltage (H = 1)
- Negative Logic (we don’t use it)
  - “0” – high voltage (H = 0)
  - “1” – low voltage (L = 1)

The figure gives the minimum voltage, called V_{SS}, and the maximum voltage, called V_{DD}, that can exist in the circuit. We will assume that V_{SS} is 0 volts, corresponding to electrical ground, denoted Gnd. The voltage V_{DD} represents the power supply voltage.



Noise Margin



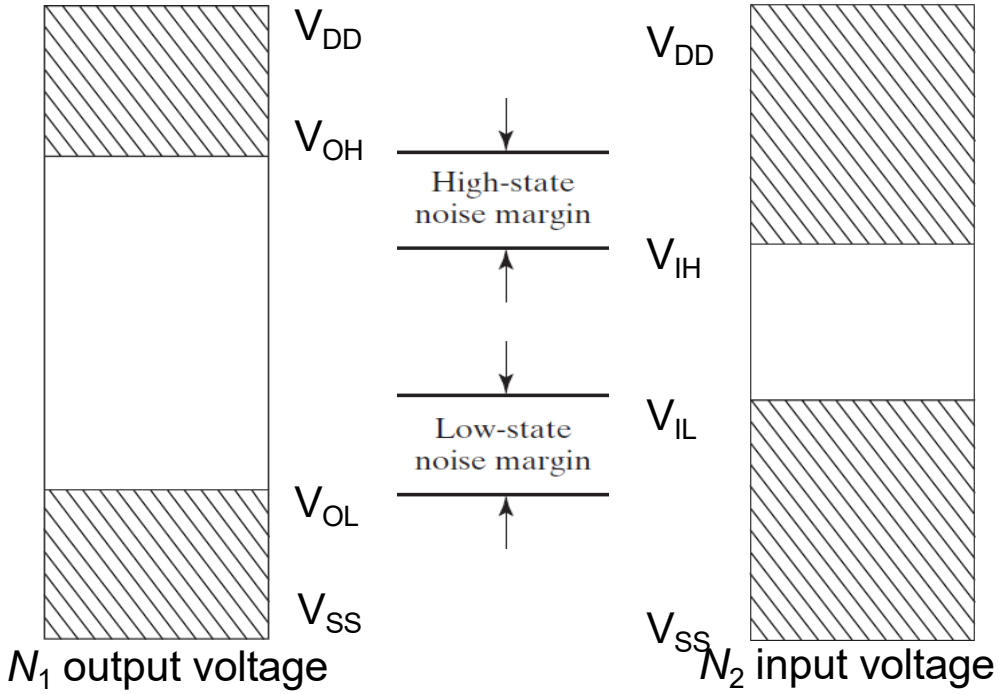
- Consider the circuit:
- Electronic circuits are subjected to random noise.
  - This can change the output voltage levels produced by gate N_1
  - This should not cause gate N_2 to misinterpret a low logic value as a high one, or vice versa.
- **Noise margin**: A measure of the extent to which a logic circuit can tolerate noise.

High state noise margin:

$NM_H = V_{OH(min)} - V_{IH\ (min)}$

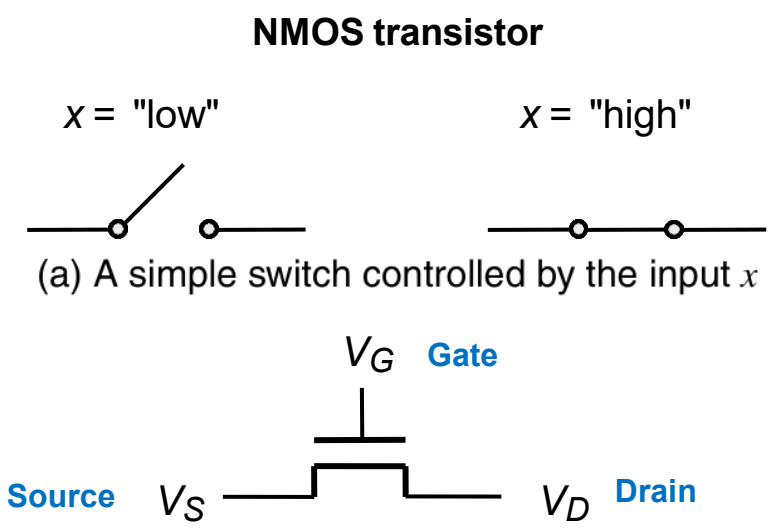
Low state noise margin

$NM_L = V_{IL(max)} - V_{OL(max)}$

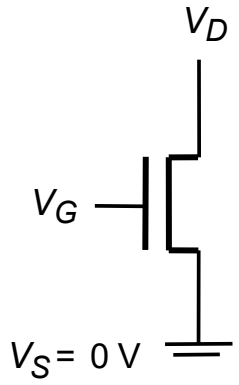


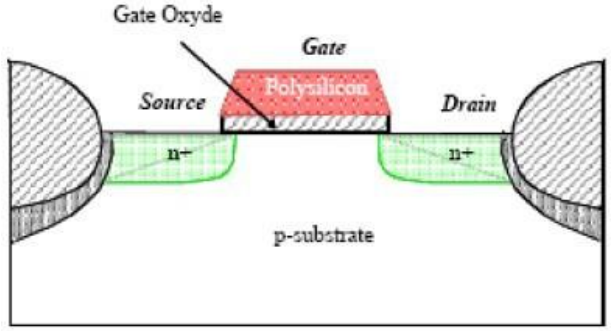
Transistor operating as switches

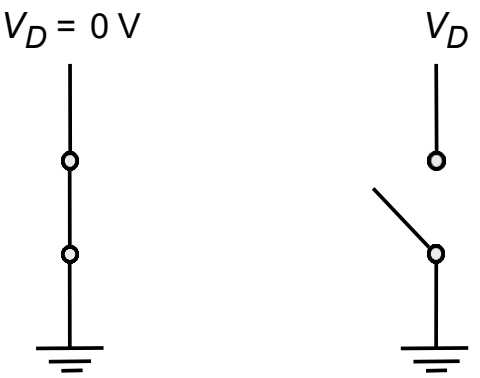
- Logic circuits are implemented using transistors (switches).
  - Full treatment in electronic courses (ELEC-252, ELEC-451)
- Most popular type of transistor: metal oxide semiconductor field-effect transistor (MOSFET). There are two different types of MOSFET transistors:
  - N-channel (NMOS) and P-channel (PMOS)



Simplified symbol for an **NMOS transistor**. It has four electrical terminals, called the source, drain, gate, and substrate.







Closed switch when $V_G = V_{DD}$

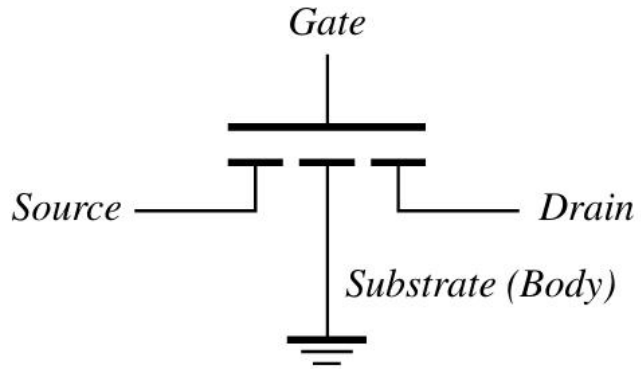
Transistor on (drain is pulled down to gnd)

Open switch when $V_G = 0$ V

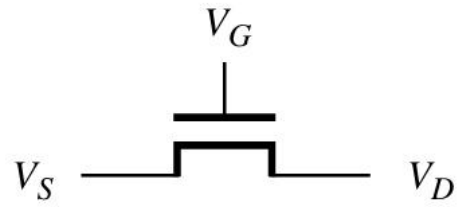
Transistor off

Transistor operating as switches NMOS transistor cont'd

- The figure gives a graphical symbol for an NMOS transistor.
- It has four electrical terminals: source, drain, gate, and substrate. The substrate (also called body) terminal is connected to Gnd.
- There is no physical difference between the source and drain terminals. They are distinguished in practice by the voltage levels applied to the transistor.
- For NMOS transistor, the terminal with the lower voltage level is deemed to be the source.



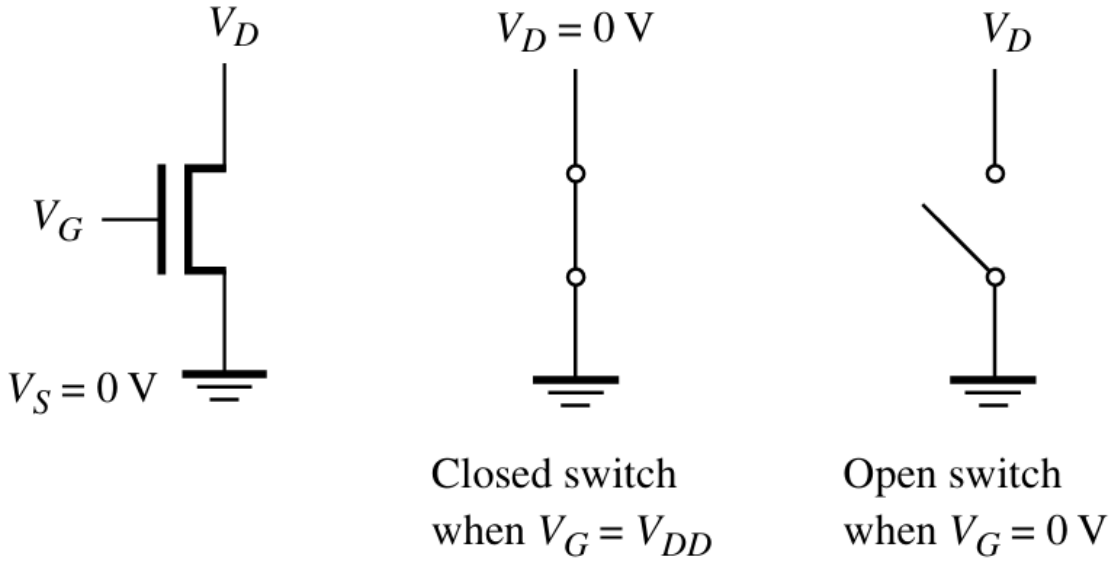
(b) NMOS transistor



(c) Simplified symbol for an NMOS transistor

Transistor operating as switches NMOS transistor cont'd

- For the NMOS transistor, the state is controlled by the voltage V_G at the gate terminal.
  - If V_G is high ($V_G = V_{DD}$), then the transistor is turned on and acts as a closed switch that connects the source and drain terminals.
  - If V_G is low, then there is no connection between the source and drain, and we say that the transistor is turned off.

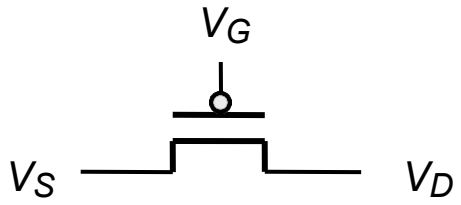


Transistor operating as switches PMOS transistor

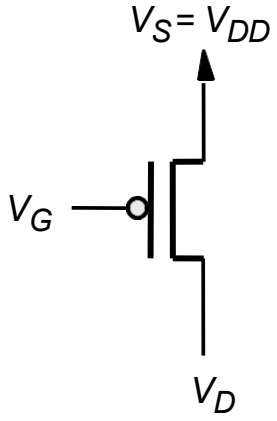
- PMOS: have the opposite behaviour of NMOS transistors.

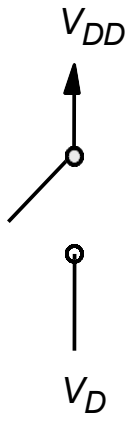


(a) A simple switch controlled by the input x



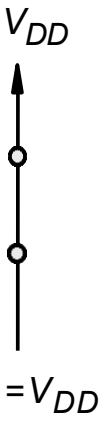
Simplified symbol for a PMOS transistor





Open switch when $V_G = V_{DD}$

Transistor off

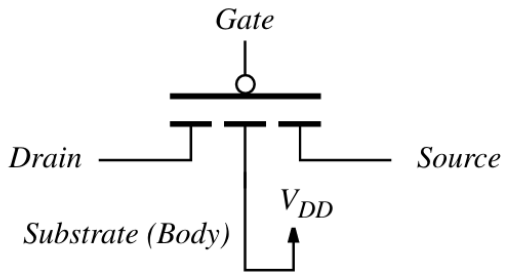


Closed switch when V_G = 0 V

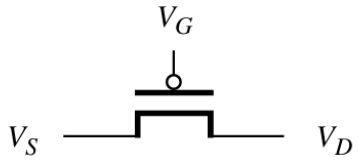
Transistor on (drain is *pulled up* to V_{DD})

Transistor operating as switches PMOS transistor

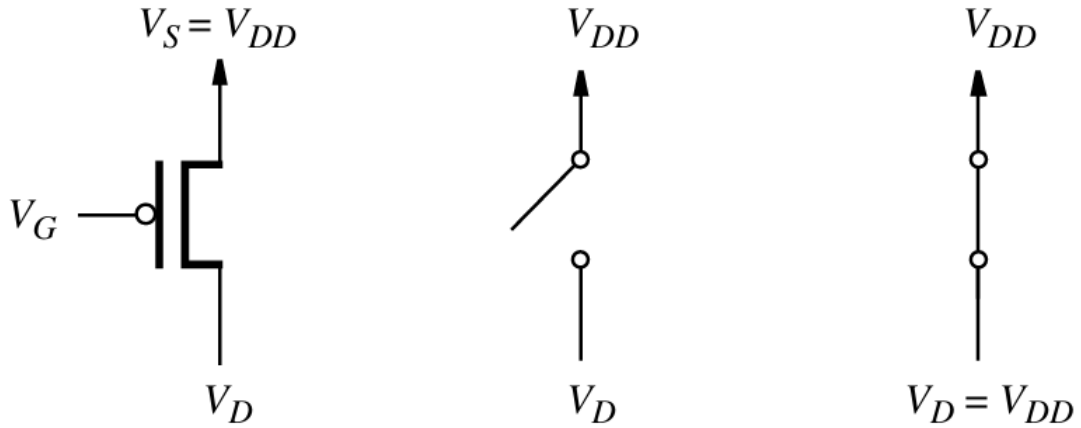
- In logic circuits the substrate of the PMOS transistor is connected to V_{DD}.
- In the PMOS transistor the **source is the node with the higher voltage**.
  - If V_G is high, then the PMOS transistor is turned off and acts like an open switch.
  - When V_G is low, the transistor is turned on and acts as a closed switch that connects the source and drain.



(b) PMOS transistor



(c) Simplified symbol for an PMOS transistor



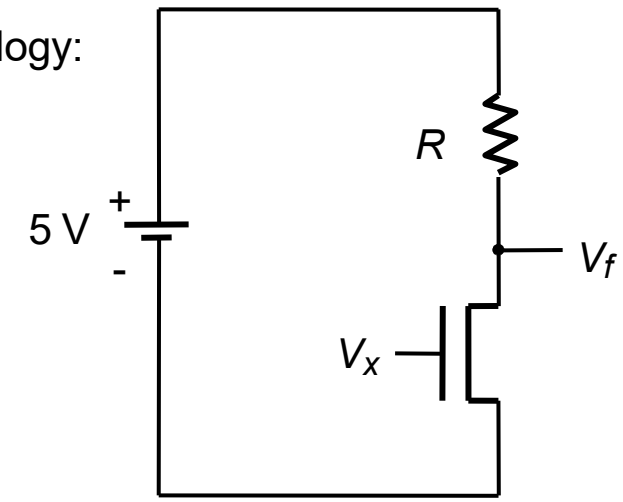
Open switch when $V_G = V_{DD}$

Closed switch when $V_G = 0$ V

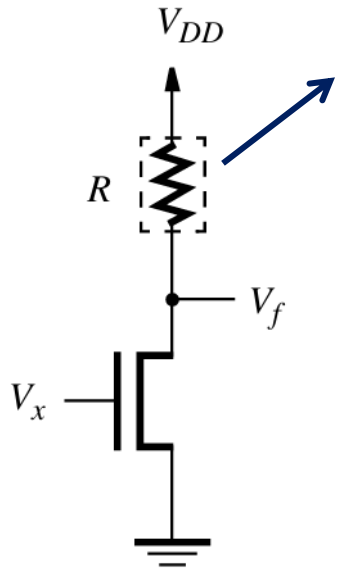
(drain is *pulled up* to V_{DD}.)

NMOS Logic Gates

- A NOT gate using NMOS technology:



Circuit diagram



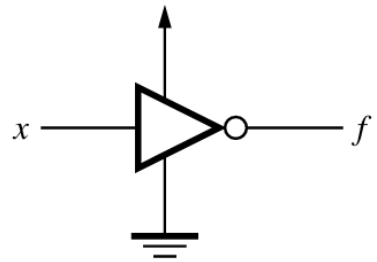
Simplified circuit diagram

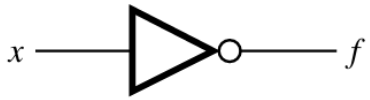
Pull-up resistor, typically implemented by a transistor.

V_x = 0 V, then V_f = 5 V

V_x = 5 V, then V_f = 0 V

$f = \overline{x}$





Graphical symbols
(NOT gate or an inverter)

The top symbol shows the input, output, power, and ground terminals, and the bottom symbol is simplified to show only the input and output terminals.

- When V_x = 0V, the transistor is turned off, no current flows through R and V_f = 5V.
- When V_x = 5V, the transistor is turned on and pulls V_f to a low voltage level.
- If V_f is viewed as a function of V_x, then the circuit is an NMOS implementation of a NOT gate. In logic terms this circuit implements the function f = $\bar{x}$.

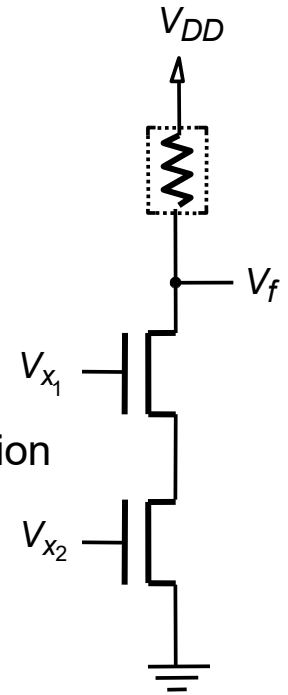
NMOS Logic Gates (Cont'd)

- A **NAND gate** using NMOS technology:

$V_{x1} = V_{x2} = 5\ V \quad V_f = 0\ V$

V_{x1} or $V_{x2} = 0\ V \quad V_f = 5\ V$

$$f = \overline{x_1 x_2}$$



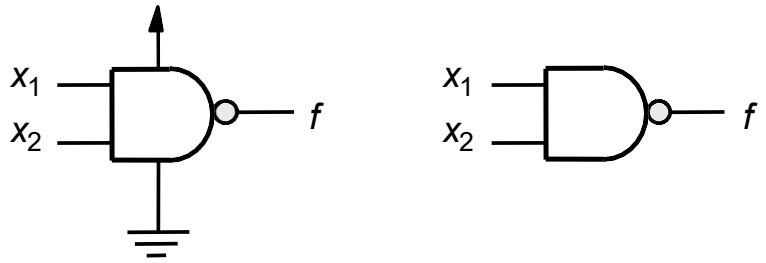
Series connection

Circuit diagram

x_1	x	f
0	0	1
0	1	1
1	0	1
1	1	0

Truth table

- Circuit Explanation:
  - If $V_{x1} = V_{x2} = 5V$, then both transistors are on $V_f = 0V$ as V_f gets shorted to gnd.
  - If either $V_{x1} = 0V$ or $V_{x2} = 0V$, that is, if either of the transistors are off, no current will flow through the series connected transistors and V_f is pulled to 5V.



Graphical symbols

NMOS Logic Gates (Cont'd)

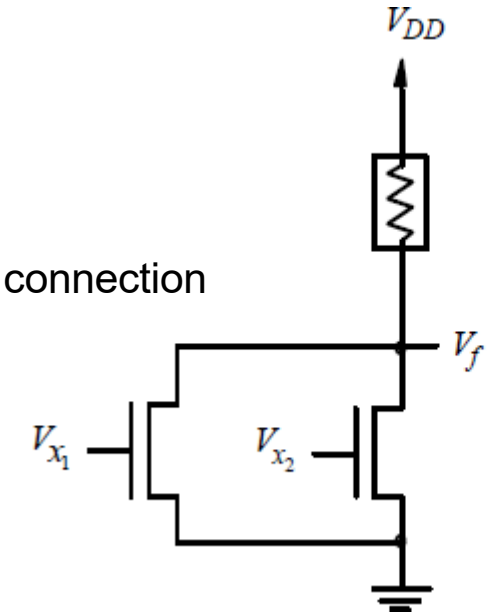
- A **NOR gate** using NMOS technology:

V_{x1} or V_{x2} = 5 V V_f = 0 V

V_{x1} = V_{x2} = 0 V V_f = 5 V

$$f = \overline{x_1 + x_2}$$

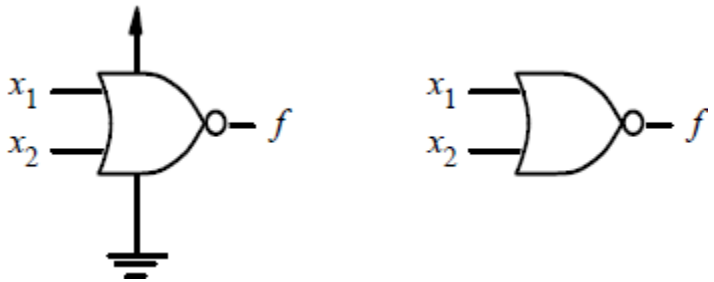
Parallel connection



x_1	x_2	f
0	0	1
0	1	0
1	0	0
1	1	0

Circuit diagram Truth table

- Circuit Explanation:
  - if either V_{x1} = 5V or V_{x2} = 5V, then V_f will be close to 0 V.
  - Only if both V_{x1} and V_{x2} are 0 will V_f be pulled up to 5 V.

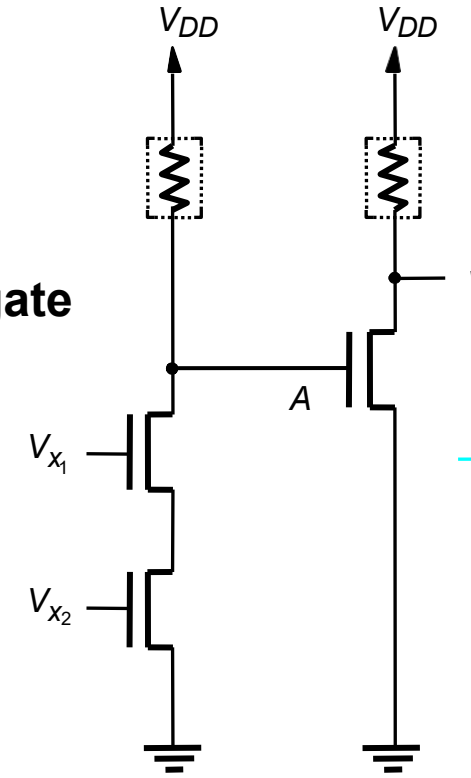


Graphical symbols

NMOS Logic Gates (Cont'd)

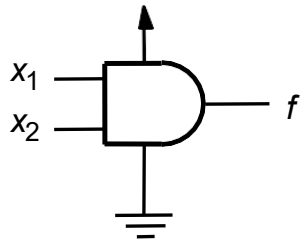
- An **AND and an OR gate** using NMOS technology:

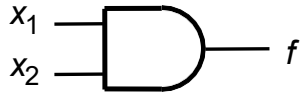
AND gate



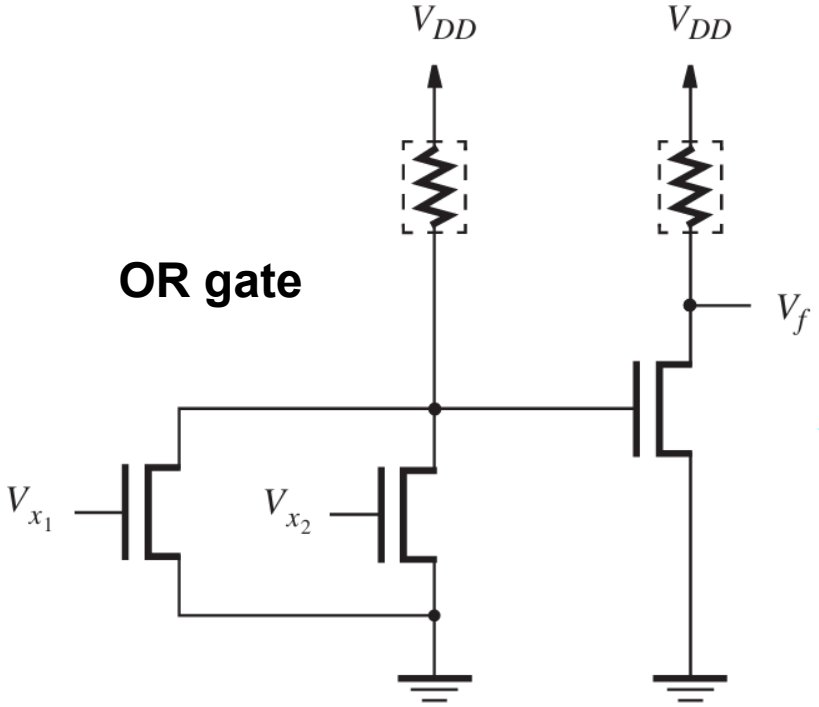
NAND Inverter

x_1	x_2	f
0	0	0
0	1	0
1	0	0
1	1	1



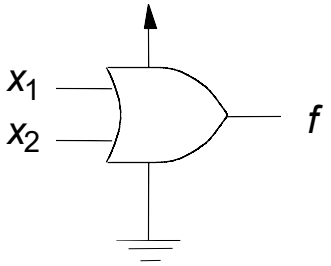


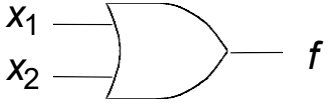
OR gate



NOR Inverter

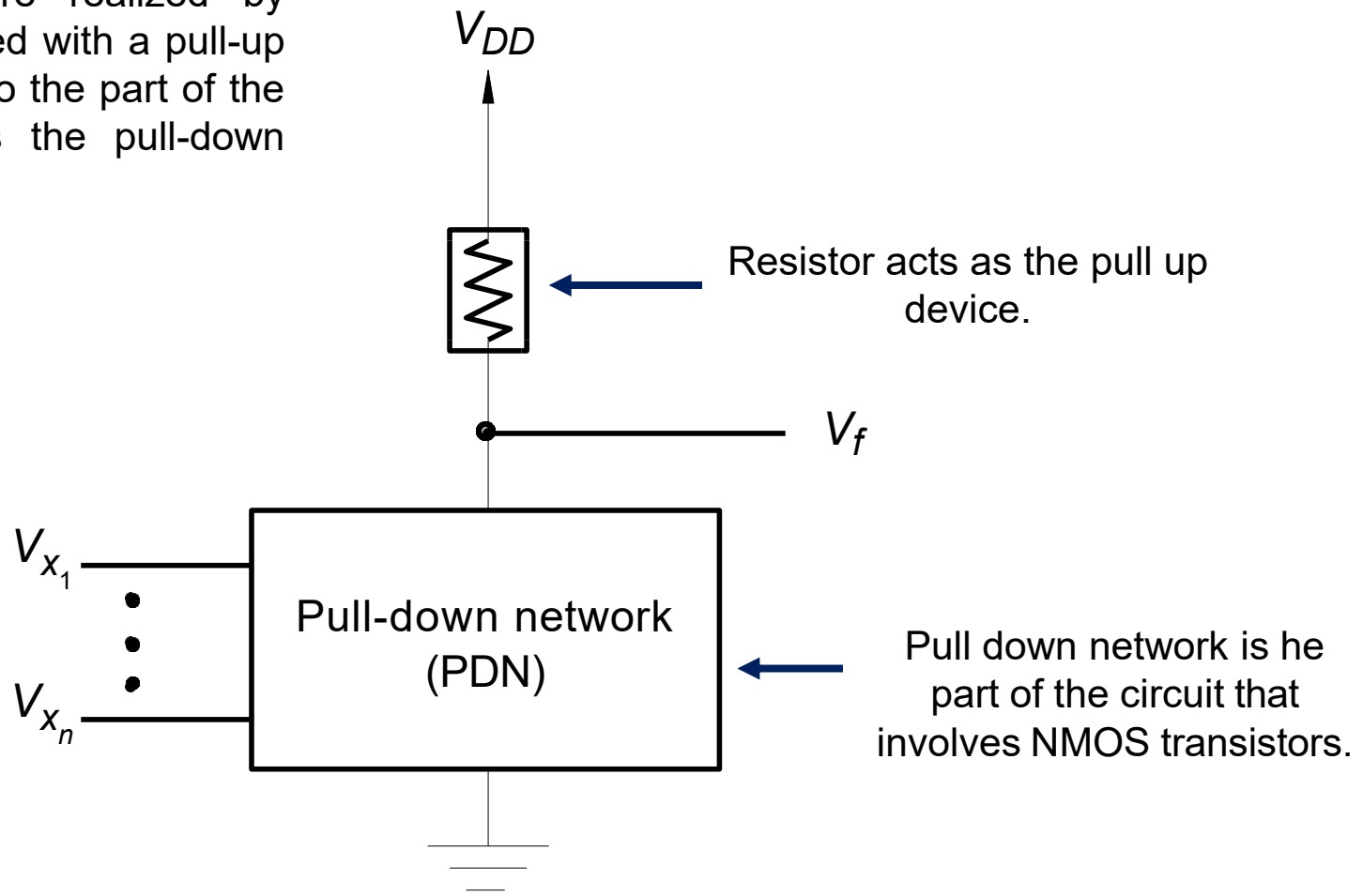
x_1	x_2	f
0	0	0
0	1	1
1	0	1
1	1	1





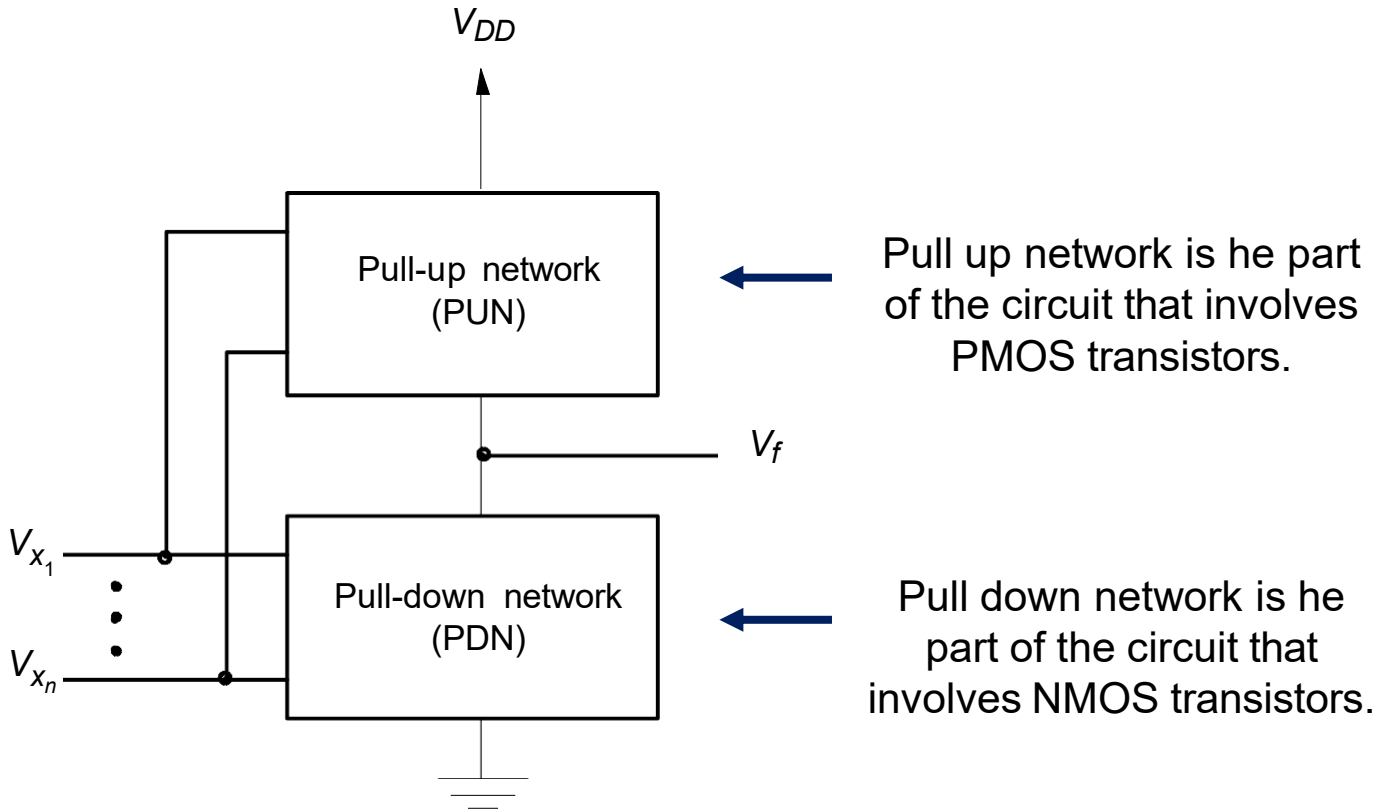
NMOS Logic Gates (Cont'd)

- In NMOS circuits the logic functions are realized by arrangements of NMOS transistors, combined with a pull-up device that acts as a resistor. We will refer to the part of the circuit that involves NMOS transistors as the pull-down network (PDN).
- Structure of an NMOS circuit:



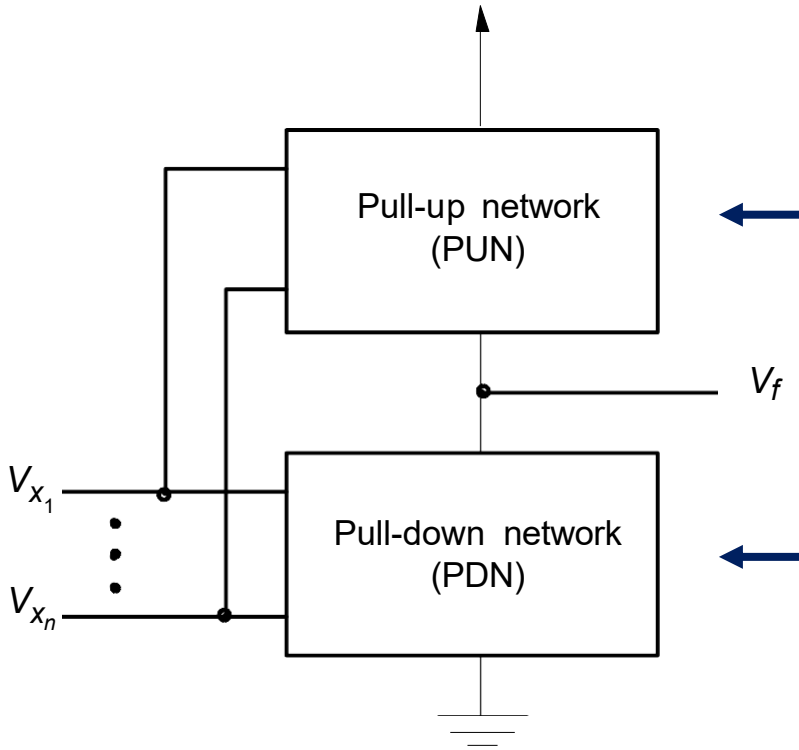
CMOS Logic Gates

- CMOS: Complementary MOS (both NMOS and PMOS):
  - Pull-up device is replaced with a pull-up network (PUN) using PMOS transistors.
  - PDN and PUN networks have complementary functions, being dual of each other, one having transistors in series, the other in parallel, and vice versa.



Structure of an CMOS circuit

CMOS Logic Gates



Structure of an CMOS circuit

Pull up network is he part of the circuit that involves PMOS transistors.

PUN pulls V_f up to V_{DD}

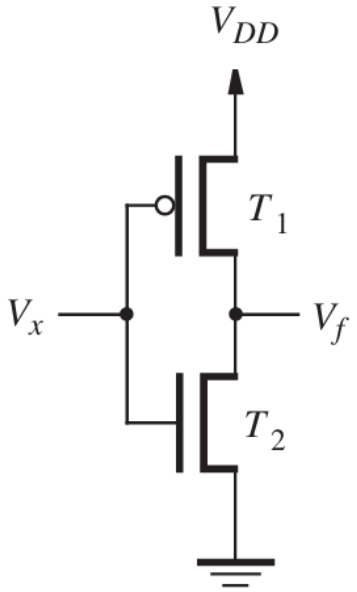
Pull down network is he part of the circuit that involves NMOS transistors.

PDN pulls V_f down to Gnd

- For any given valuation of the input signals, either the **PDN pulls V_f down to Gnd** or the **PUN pulls V_f up to V_{DD}.**
- The PDN and the PUN have equal numbers of transistors, which are arranged so that the two networks are duals of one another. Wherever the PDN has NMOS transistors in series, the PUN has PMOS transistors in parallel, and vice versa.V_{DD}

CMOS Logic Gates (Cont'd)

- **A CMOS NOT gate:**
  - When Vx = 0V, transistor T2 is off and transistor T1 is on. This makes Vf = 5 V, and since T2 is off, no current flows through the transistors.
  - When Vx = 5V,T2 is on and T1 is off. Thus Vf = 0V,and no current flows because T1 is off.



(a) Circuit

$f = \overline{x}$

x	T_1	T_2	f
0	on	off	1
1	off	on	0

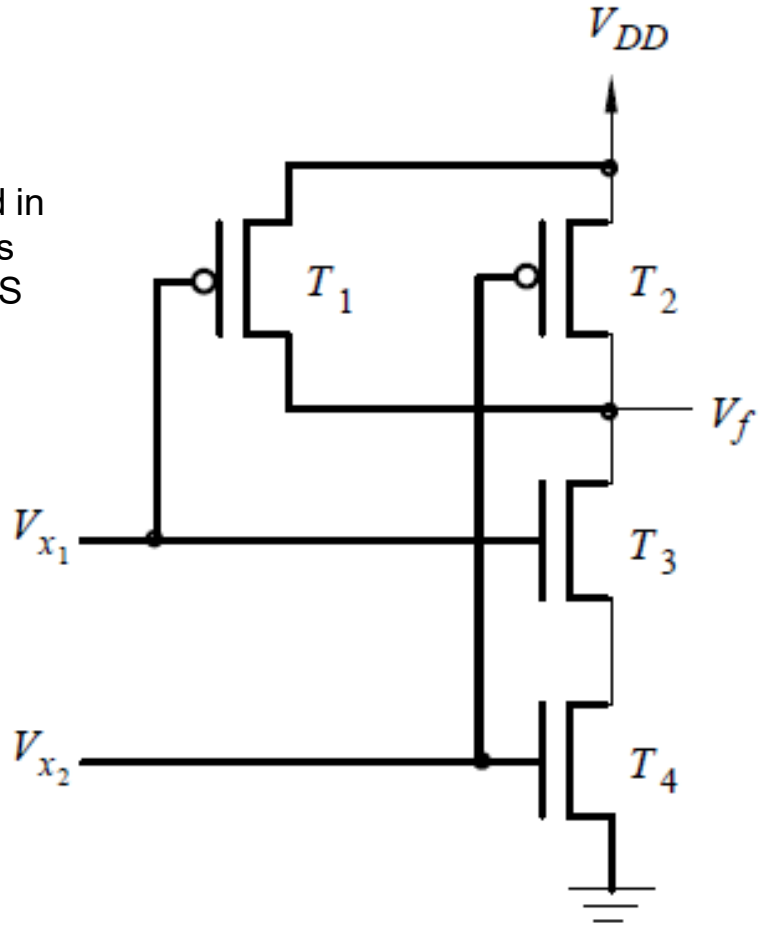
(b) Truth table and transistor states

- CMOS is popular because no power is dissipated under steady state conditions (no current flows when the input is either low or high).
- Still having leakage current
- Dynamic power consumption

CMOS Logic Gates (Cont'd)

- A CMOS **NAND** gate:

It is similar to the NMOS circuit presented in slide 12 except that the pull-up device has been replaced by the PUN with two PMOS transistors connected in parallel.



Circuit

$$f = \overline{x_1 x_2}$$

x_1	x_2	T_1	T_2	T_3	T_4	f
0	0	on	on	off	off	1
0	1	on	off	off	on	1
1	0	off	on	on	off	1
1	1	off	off	on	on	0

Truth table and transistor states

Explanation of CMOS NAND gate circuit

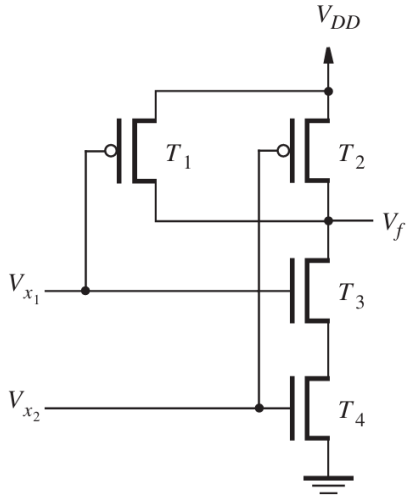
The circuit in Figure B.13 can be derived from the logic expression that defines the NAND operation, $f = \overline{x_1 x_2}$. This expression specifies the conditions for which $f = 1$; hence it defines the PUN. Since the PUN consists of PMOS transistors, which are turned on when their control (gate) inputs are set to 0, an input variable x_i turns on a transistor if $x_i = 0$. From DeMorgan's law, we have

$$f = \overline{x_1 x_2} = \bar{x}_1 + \bar{x}_2$$

Thus $f = 1$ when *either* input x_1 or x_2 has the value 0, which means that the PUN must have two PMOS transistors connected in parallel. The PDN must implement the complement of f, which is

$$\bar{f} = x_1 x_2$$

Since $\bar{f} = 1$ when *both* x_1 and x_2 are 1, it follows that the PDN must have two NMOS transistors connected in series.



(a) Circuit

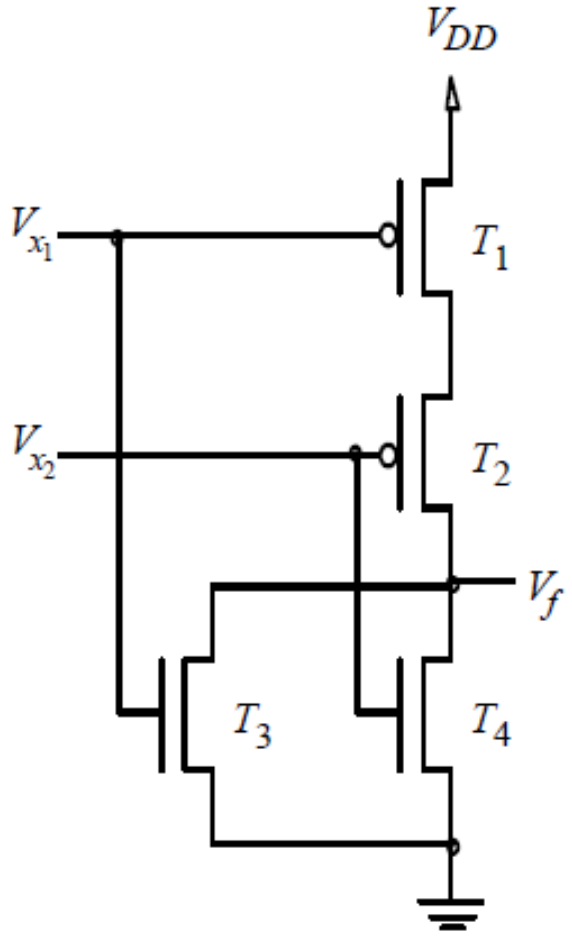
x_1	x_2	T_1	T_2	T_3	T_4	f
0	0	on	on	off	off	1
0	1	on	off	off	on	1
1	0	off	on	on	off	1
1	1	off	off	on	on	0

(b) Truth table and transistor states

Figure B.13 CMOS realization of a NAND gate.

CMOS Logic Gates (Cont'd)

- A CMOS **NOR** gate:



$$f = \overline{x_1 + x_2}$$

x_1	x_2	T_1	T_2	T_3	T_4	f
0	0	on	on	off	off	1
0	1	on	off	off	on	0
1	0	off	on	on	off	0
1	1	off	off	on	on	0

Circuit

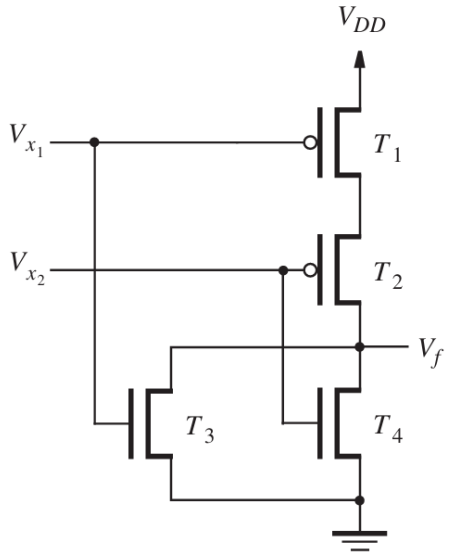
Truth table and transistor states

Explanation of CMOS NOR gate circuit

The circuit for a CMOS NOR gate is derived from the logic expression that defines the NOR operation

$$f = \overline{x_1 + x_2} = \bar{x}_1 \bar{x}_2$$

Since $f = 1$ only if both x_1 and x_2 have the value 0, then the PUN consists of two PMOS transistors connected in series. The PDN, which realizes $\bar{f} = x_1 + x_2$, has two NMOS transistors in parallel, leading to the circuit shown in Figure B.14.



(a) Circuit

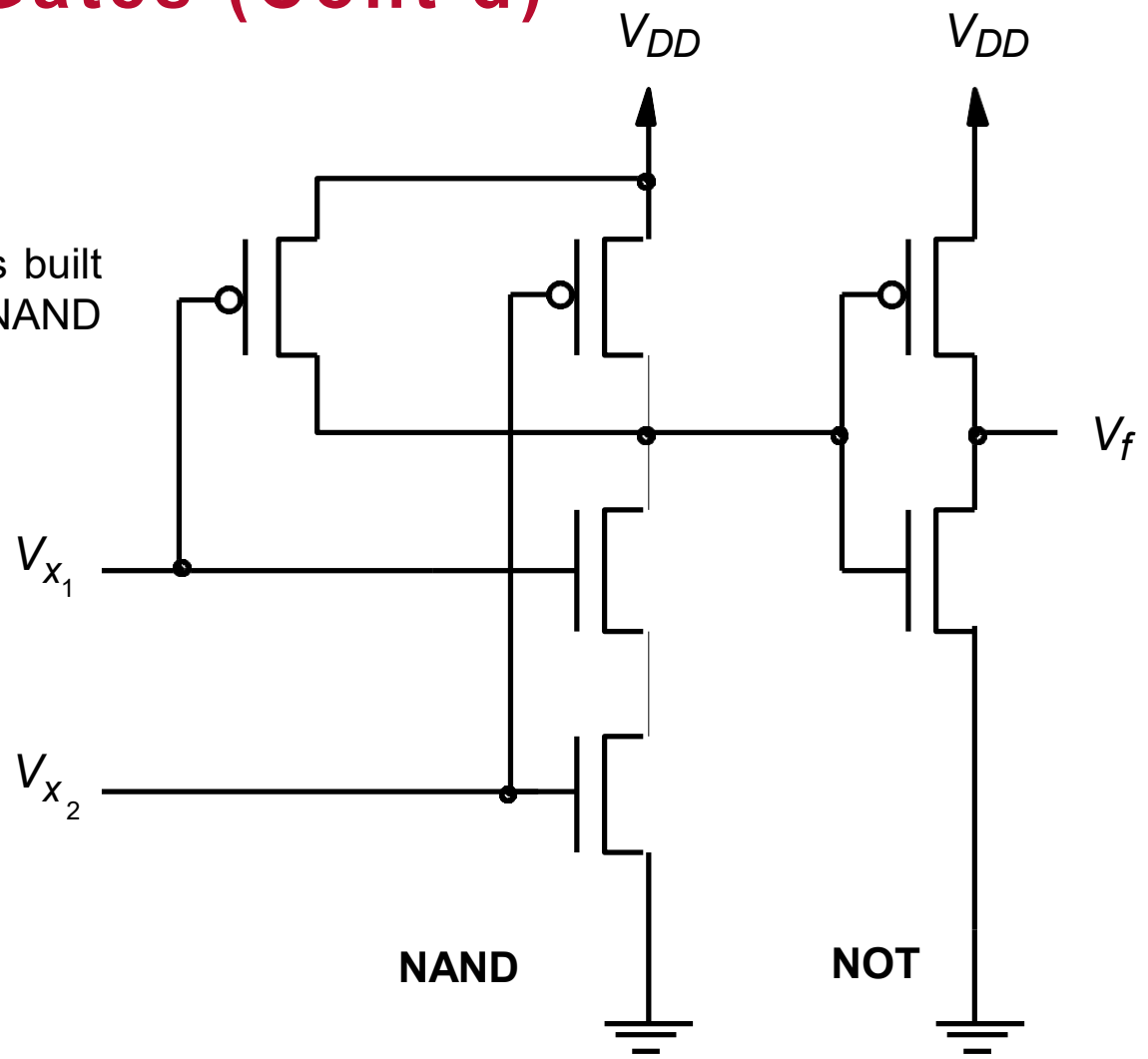
x_1	x_2	T_1	T_2	T_3	T_4	f
0	0	on	on	off	off	1
0	1	on	off	off	on	0
1	0	off	on	on	off	0
1	1	off	off	on	on	0

(b) Truth table and transistor states

Figure B.14 CMOS realization of a NOR gate.

CMOS Logic Gates (Cont'd)

- A CMOS **AND** gate:
- A CMOS AND gate is built by connecting a NAND gate to an inverter



$$f = \overline{\overline{x_1 x_2}} = x_1 x_2$$

Similarly, an OR can be obtained from a NOR followed by a NOT.

CMOS Logic Gates (Cont'd)

Example B.1

Consider the function

Example B.1

$$f = \bar{x}_1 + \bar{x}_2\bar{x}_3$$

Since all variables appear in their complemented form, we can directly derive the PUN. It consists of a PMOS transistor controlled by x_1 in parallel with a series combination of PMOS transistors controlled by x_2 and x_3. For the PDN we have

$$\bar{f} = \overline{\bar{x}_1 + \bar{x}_2\bar{x}_3} = x_1(x_2 + x_3)$$

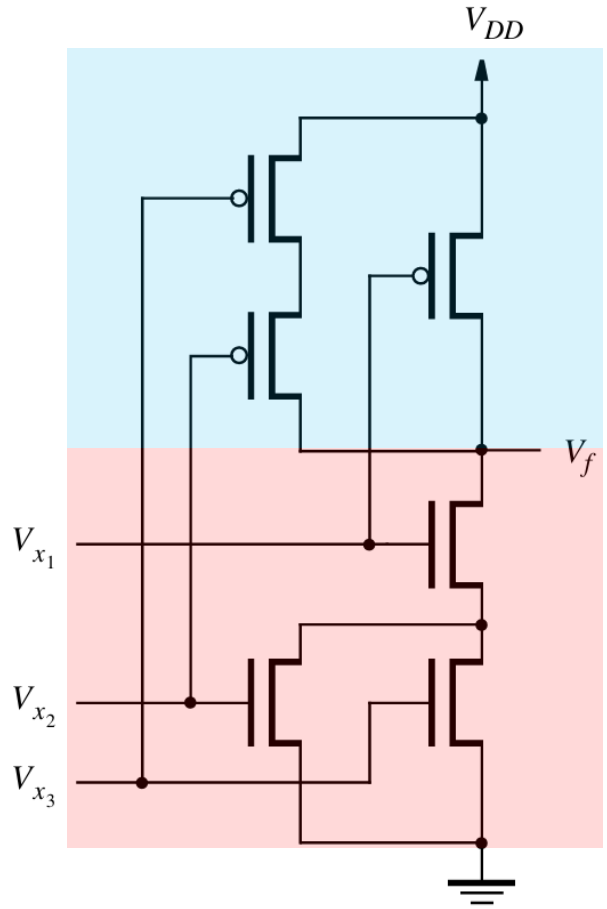
This expression gives the PDN that has an NMOS transistor controlled by x_1 in series with a parallel combination of NMOS transistors controlled by x_2 and x_3. The circuit is shown in Figure B.16.

$$f = \bar{x}_1 + \bar{x}_2\bar{x}_3$$

PUN
(implements f)

$$\bar{f} = \overline{\bar{x}_1 + \bar{x}_2\bar{x}_3} = x_1(x_2 + x_3)$$

PDN
(implements $\bar{f}$)



CMOS Logic Gates (Cont'd)

Example B.2 Consider the function

$$f = \bar{x}_1 + (\bar{x}_2 + \bar{x}_3)\bar{x}_4$$

Then

$$\bar{f} = x_1(x_2x_3 + x_4)$$

These expressions lead directly to the circuit in Figure B.17.

PUN
(implements f)

$f = \bar{x}_1 + (\bar{x}_2 + \bar{x}_3)\bar{x}_4$

PDN
(implements $\bar{f}$)

$\bar{f} = x_1(x_2x_3 + x_4)$

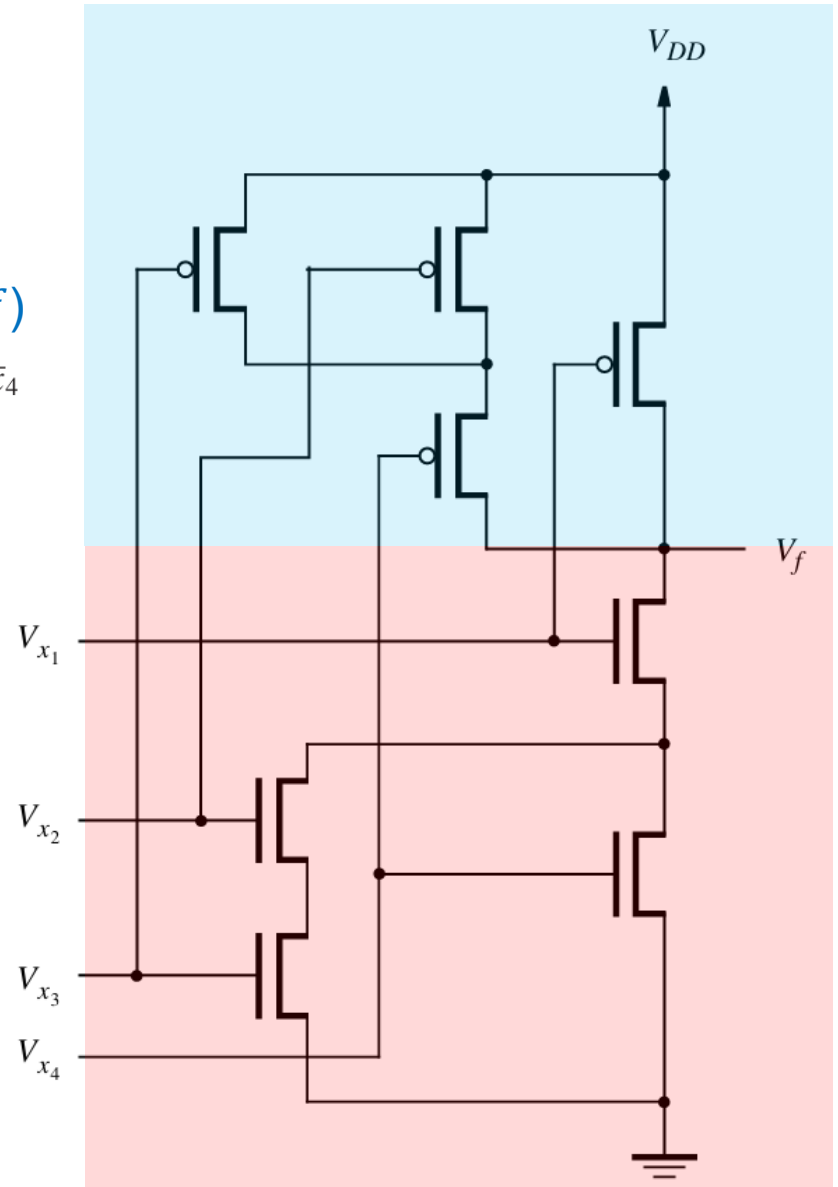


Figure B.17 The circuit for Example B.2.


Queen's
UNIVERSITY